Vim Notes ■

Vim is a modal editor, which means its behavior depends on the mode you are in.
The main modes are:

- Normal Mode → default mode (navigate, delete, copy, paste, etc.)
- Insert Mode → for typing text (like a regular editor)
- Visual Mode → for selecting text
- Command-line Mode → for running commands (save, quit, search, etc.)

1. Opening and Quitting

```
vim file.txt        # open a file

Inside Vim:
:q      → quit (fails if changes unsaved)
:q!     → quit without saving
:w      → save (write) file
:wq     → save and quit
:x      → save and quit
ZZ      → save and quit (shortcut, no colon needed)
```

2. Modes in Vim

```
Normal Mode:
- Default mode when Vim starts.
- Used for navigation, editing commands.
- Enter from any mode with Esc.

Insert Mode:
- For typing text directly.
- Enter with:
    i   → insert before cursor
    I   → insert at start of line
    a   → append (insert after cursor)
    A   → append at end of line
    o   → open new line below
    O   → open new line above
- Exit with Esc.

Visual Mode:
- For selecting text.
    v       → character-wise select
    V       → line-wise select
    Ctrl-v  → block (column) select

Command-line Mode:
- For commands like :w, :q, :s, /pattern.
- Enter with :
```

3. Moving the Cursor

```
In Normal Mode:
h → left
l → right
```

```
j → down
k → up

Word motions:
w → start of next word
e → end of word
b → beginning of word

Line motions:
0 → beginning of line
^ → first non-space character
$ → end of line

File motions:
gg  → top of file
G   → bottom of file
:n  → go to line n (example: :15)

In Insert Mode:
- Use Arrow keys or Enter to move around.
- Backspace works normally.
```

4. Editing Text

```
From Normal Mode:
x   → delete character under cursor
dw  → delete word
dd  → delete entire line
yy  → copy (yank) line
p   → paste after cursor
u   → undo
Ctrl-r → redo

Example:
File content:
Hello World

Command in Normal Mode: dd
Result:
<line removed>
```

5. Searching and Replacing

```
/word   → search forward
?word   → search backward
n       → next match
N       → previous match

Replace:
:s/old/new/     → replace first old with new on current line
:s/old/new/g    → replace all on current line
:%s/old/new/g   → replace in entire file
```

6. Combining Motions with Operators

```
In Vim, operator + motion = action.

Examples:
d$  → delete until end of line
y0  → yank until beginning of line
caw → change a word (delete + go into insert mode)
```

7. Example Workflow

```
vim notes.txt

Inside Vim:
1. Press i → type Hello.
2. Press Enter → new line, type World.
3. Press Esc → back to Normal Mode.
4. Type dd → deletes World.
5. Type :wq → save & quit.

File now contains:
Hello
```

8. Enabling Mouse Support (Optional)

```
By default, mouse is off. To enable temporarily:
:set mouse=a

To enable always, add to ~/.vimrc:
set mouse=a
```

Quick Summary

```
Modes: Normal (Esc), Insert (i), Visual (v), Command (:).
Move: h j k l, w, e, b, gg, G.
Edit: x, dd, yy, p, u, Ctrl-r.
Search/Replace: /, ?, :s, :%s.
Save/Quit: :w, :q, :wq, ZZ.
```